6 Object detection

- in Chapter 3, we studied various convolutional neural networks that are commonly used in computer vision, and applied them to simple image classification tasks
- at the beginning of this chapter, we will describe two methods that may improve model generalization, namely *image augmentation* and *fine-tuning*
- since deep neural networks can effectively represent images in multiple levels, such layer-wise representations have been successfully used in various computer vision tasks, such as *object detection* and *semantic segmentation*
- following the key idea of leveraging layer-wise representations in computer vision, we will begin with major components and techniques for object detection